

Linear Algebra—Winter/Spring 2026

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End-Semester Examination

Wednesday, April 29, 2026

3:30 PM to 6:30 PM

Instructions

- i. *Studied beyond the course?* If you use any concept that we have not covered in this course (e.g. relations between determinant/trace and eigenvalues), please be sure to first define the appropriate objects and prove any property thereof, that you use in your answers.
 - ii. *Duration.* You have a total of 3 hours.
 - iii. *Do I need to answer all the questions?* There are three types of problems. Each type has options. This is written explicitly inline.
 - iv. *Points distribution.*
 - You earn 10 points for an A-Type problem, and 16 points for a B-Type problem.
 - You can score a maximum of $4 \times 10 + 3 \times 16 = 88$ points by correctly solving 4 A-Type and 3 B-Type problems.
 - C-Type problems will contribute to the assignment scaling score.
 - v. *What if I answer more than the required number of questions?*
 - If, for any given type, you are asked to answer k questions, and you write $k' > k$ answers, then only the first k answers (of that type) will be graded.
 - You are therefore encouraged to only submit k answers (and strike out the rest, should you attempt more than k answers).
 - vi. *About Hints.*
 - Note that there may be more than one way of solving a question so do not obsess over hints.
 - Additionally, we have included a set of facts at the end of this document that you may find useful.
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A Type — do 4 out of 7 (10 pts each)

1. Let V be a finite-dimensional vector space and let T be a linear operator on V . Suppose that $\text{rank}(T^2) = \text{rank}(T)$. Prove that the range and null space of T are disjoint, i.e., have only the zero vector 0 in common.

Hint: You can use the rank-nullity without proof, if needed.

2. Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is normal, i.e., $TT^* = T^*T$, and $T(1, 1, 1) = (2, 2, 2)$. Suppose $(z_1, z_2, z_3) \in \text{null } T$ (the null space of T). Prove that $z_1 + z_2 + z_3 = 0$.

Hint: You may use the facts at the end, without proof.

3. Fix $u, x \in V$. Define $T \in \mathcal{L}(V)$ by

$$Tv = \langle v, u \rangle x, \text{ for every } v \in V.$$

Suppose the underlying field is $\mathbb{F} = \mathbb{R}$. Prove that T is self-adjoint if and only if u and x are linearly dependent. Does this result remain true if $\mathbb{F} = \mathbb{C}$? Justify.

4. (a) Let $n \geq 3$. Let $A = [A_{ij}]$ be an $n \times n$ matrix with entries $A_{ij} = i + j$. Find $\det(A)$.

(b) Let $n \geq 2$. Let $B = [B_{ij}]$ be an $n \times n$ matrix with entries $B_{ij} = ij$. Find $\det(B)$.

Hint: You may use the properties of the determinant instead of the explicit formula.

5. Consider the operator $T \in \mathcal{L}(\mathbb{F}^5)$ that acts as

$$T(x, y, z, u, v) = (x, y + 2z + 3u + 4v, 3z + 2u + 3v, 4u + 2v, 5v).$$

Prove that there exists a $\lambda \in \mathbb{F}$ and a 2-dimensional subspace U of $\mathbb{F}^5$ such that

- U is invariant under T and
- for every $u \in U$, $Tu = \lambda u$.

Note that proving existence is enough.

Hint: This question requires almost no computation to answer.

What is the matrix representation of T ? Maybe focus on the first row and column; and then focus on the rest of the matrix.

You can use theorem statements from the text, without proof, in your argument.

6. Let V be finite-dimensional (over the field $\mathbb{F}$) and $\phi : V \rightarrow \mathbb{F}$ be a linear functional on V . Show that there is some vector $u \in V$ such that $\phi(v) = \langle v, u \rangle$ for all $v \in V$ (no need to show uniqueness).

Hint: Start by writing v as a linear combination of standard basis vectors, substitute it in $\phi(v)$ and use the linearity of ϕ .

7. Suppose $0 < \theta < \pi/2$. Demonstrate the Gram-Schmidt procedure on the following vectors (in $\mathbb{R}^2$)

$$(\cos \theta, \sin \theta) \text{ and } (0, 1)$$

by proceeding in two different ways:

$$\begin{aligned} & \langle T u, v \rangle = \langle u, T^* v \rangle \\ & \langle T u, v \rangle = \langle u, T^* v \rangle \end{aligned}$$

(a) Treat $(0, 1)$ as the first vector.

(b) Treat $(\cos \theta, \sin \theta)$ as the first vector.

By the first vector, it is meant that this vector should be part of the final orthonormal basis.

B Type — do 3 out of 4 (16 pts each)

8. Let T be the linear operator on $\mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

(a) Find matrix of T with respect to the ordered standard basis $\{e_1, e_2, e_3\}$.

(b) Obtain the row reduced echelon form of the matrix of T (in the standard basis). Show the calculation explicitly.

(c) Using the calculation in (b), find the matrix for T^{-1} (in the standard basis).

9. Suppose $T \in \mathcal{L}(V)$ has a singular value decomposition given by

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \cdots + s_n \langle v, e_n \rangle f_n$$

for every $v \in V$, where $s_1, s_2, \dots, s_n$ are strictly positive singular values of T and $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are orthonormal bases of V .

(a) Prove that if $v \in V$, then

$$T^*Tv = s_1 \langle v, f_1 \rangle e_1 + \cdots + s_n \langle v, f_n \rangle e_n.$$

(b) Prove that if $v \in V$, then

$$T^*Tv = s_1^2 \langle v, e_1 \rangle e_1 + \cdots + s_n^2 \langle v, e_n \rangle e_n.$$

(c) Suppose $S \in \mathcal{L}(V)$ is defined by, for every $v \in V$,

$$Sv = \frac{\langle v, f_1 \rangle e_1}{s_1} + \cdots + \frac{\langle v, f_n \rangle e_n}{s_n}.$$

Prove that ST is the identity operator on V .

10. Suppose $T \in \mathcal{L}(V)$ is a linear operator on the vector space V . Show that for each i , U_i (as described below) is an invariant subspace of V .

(a) Let T and V be arbitrary and define $U_1 := \text{nullspace}(T)$, $U_2 := \text{range}(T)$. Now suppose T has an eigenvalue λ and define $U_3 := \text{range}(T - \lambda I)$.

Hint: It may help to write Tu in terms of $(T - \lambda I)u$.

- atmos?

Suppose $u, v \in V$ where V is an inner product space. Then,

and that equality holds if and only if $\langle u, v \rangle = \|u\| \cdot \|v\|$.

- (b) Show that

$$\leq \|u\|^2 + \|v\|^2 + 2 \cdot \|u\| \cdot \|v\|. \quad (3)$$

(c) It is clear that Inequality (1) is saturated (i.e. the inequality becomes an equality), if and only if, both inequalities (2) and (3) hold and are saturated.

Hint: You can use this fact (without proof): $|\alpha| = \Re(\alpha)$ implies $\alpha = |\alpha|$, for any complex number α .

12. Let V and W be vector spaces over the field F and let T be a linear transformation from V to W . Suppose the vectors $\{\alpha_1 \dots \alpha_k\}$ form a basis for the null space of T . Let $\{\alpha_{k+1}, \dots, \alpha_n\}$ be vectors such that $\{\alpha_1, \dots, \alpha_n\}$ are a basis for V . Show that $\{T\alpha_{k+1}, \dots, T\alpha_n\}$ are linearly independent and span the range of T .

- $$\mathcal{M}(T, (e_1, \dots, e_n)) = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ 0 & a_{2,2} & a_{2,3} & \dots & a_{2,n} \\ 0 & 0 & a_{3,3} & \ddots & \\ 0 & 0 & 0 & \dots & a_{n,n} \end{bmatrix}.$$

Suppose $a_{1,2} = 0$. Prove that $a_{2,3} = \cdots = a_{2,n} = 0$.

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14. Let m, n be positive integers. Suppose R is an $m \times n$ row-reduced echelon matrix over some field F and denote its non-zero row vectors by $\rho_1, \rho_2 \dots \rho_r$. Let W be its row space and suppose the vector $\beta = (b_1, b_2 \dots b_n)$ is in W . Below, we use k_i to denote the index of the first non-zero column in the row vector ρ_i .

- i. Find the coefficients $c_1, \dots, c_r$ (in terms of $b_1, \dots, b_n$) when β is written as $\beta = c_1 \rho_1 + \dots + c_r \rho_r$.
- ii. Let $1 \leq s \leq r$. Prove that there is exactly one vector σ_s in W such that the k_s th coordinate of $\sigma_s = 1$ and the remaining k_i th coordinates (for $i \in \{1 \dots r\} \setminus s$) of σ_s are zero.
- iii. Is $\sigma_s = \rho_s$? Prove your answer.

Facts for reference

Potentially useful definitions.

- *Null space and range of T .*

For $T \in \mathcal{L}(V, W)$, the *null space* of T is given by

$$\text{null}(T) := \{v \in V : Tv = 0\}$$

and the *range* of T is given by

$$\text{range}(T) := \{w \in W : Tv = w \text{ for some } v \in V\}.$$

- *Eigenvector/eigenvalue/eigenspace.*

Suppose $T \in \mathcal{L}(V)$. A number $\lambda \in \mathbb{F}$ is called an *eigenvalue* of T if there exists a $v \in V$ such that $v \neq 0$ and $Tv = \lambda v$. Here, v is an *eigenvector* corresponding to the eigenvalue λ .

The subspace $E(\lambda, T) := \text{nullspace}(T - \lambda I) = \{v \in V : Tv = \lambda v\}$ is called the *eigenspace* of T corresponding to λ .

- *Adjoint & Self-Adjoint.*

The *adjoint* of $T \in \mathcal{L}(V)$ is defined to be the operator $T^* \in \mathcal{L}(V)$ such that $\langle Tv, w \rangle = \langle v, T^*w \rangle$, for all $v, w \in V$.

An operator $T \in \mathcal{L}(V)$ is called *self-adjoint* if $T^* = T$, i.e.,

$$\langle Tv, w \rangle = \langle v, Tw \rangle, \quad \forall v, w \in V.$$

- *Standard basis.* For instance, in $\mathbb{F}^3$, the standard basis is given by $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$.

Potentially useful results.

- *Rank-Nullity Theorem.* For $T \in \mathcal{L}(V, W)$, where V is finite-dimensional,

$$\dim V = \dim \text{null } T + \dim \text{range } T.$$

- Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V . Then, the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.
- *Schur's Theorem.* Suppose V is a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Then T has an upper-triangular matrix with respect to some orthonormal basis of V .
- *Cauchy-Schwarz Inequality.* Suppose $u, v \in V$. Then $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. This inequality is an equality if and only if one of u, v is a scalar multiple of the other.
- Suppose T is a self-adjoint operator on V such that $\langle Tv, v \rangle = 0$ for all $v \in V$. Then $T = 0$.
- An operator $T \in \mathcal{L}(V)$ is normal if and only if $\|Tv\| = \|T^*v\|$ for all $v \in V$.
- Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.